

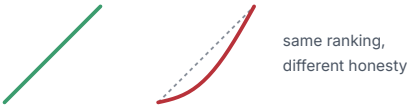
Calibration: when 0.8 really means 80%

1 Calibration = honesty

A calibrated model's 80% happens about 80% of the time. You check it by grouping and counting.

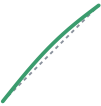
2 Ranking ≠ honesty

Two models with the same AUC (0.89) can have ECE 0.008 and 0.27.



3 Read the reliability diagram

Said vs happened, bin by bin. On the diagonal is honest; below it is overconfident. Check the counts.



4 Rebalancing breaks it

Train on 50/50 data and the model thinks attacks are common: its average P is 29% when the truth is 8%.

5 Fix it on held-out data

Platt shifts and stretches; temperature only stretches; isotonic bends. Fit on data the model never trained on.

Platt

Temperature

Isotonic

6 Check every slice

Honest on average can still be dishonest for Email, or for EDR. Check the groups you'll act on.